



Assignment No. 4

1. Write Maxwell's equation in differential & integral form for
- a) Time varying field
 - b) Free space
 - c) Good Conductor
 - d) static field

→ a) Time varying field

• Point form:-

1. $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$... derived from Faraday's law

2. $\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$... derived from ACL

3. $\rho_v = \nabla \cdot \vec{D}$... derived from Gauss law

4. $\nabla \cdot \vec{B} = 0$... derived from Gauss law

• Integral form:-

1. $\oint \vec{E} \cdot d\vec{l} = - \int_s \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$... derived from Faraday's law

2. $\oint \vec{H} \cdot d\vec{l} = \int_s \left[\vec{J} + \frac{\partial \vec{D}}{\partial t} \right] \cdot d\vec{s}$... ACL

3. $\oint_s \vec{D} \cdot d\vec{s} = \int_v \rho_v dv$... from Gauss law

4. $\oint_s \vec{B} \cdot d\vec{s} = 0$... from Gauss law



b) Free Space

$$J_v = 0 \text{ \& } \sigma = 0$$

• Point form:-

$$1. \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$2. \nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

$$3. \nabla \cdot \vec{D} = 0$$

$$4. \nabla \cdot \vec{B} = 0$$

• Integral form:-

$$1. \oint \vec{E} \cdot d\vec{l} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

$$2. \oint \vec{H} \cdot d\vec{l} = \int_S \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S}$$

$$3. \oint \vec{D} \cdot d\vec{S} = 0$$

$$4. \oint \vec{B} \cdot d\vec{S} = 0$$

c) Good Conductor

$$\vec{J} \gg \frac{\partial \vec{D}}{\partial t} \text{ \& } J_v = 0$$

• Point form:-

$$1. \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$2. \nabla \times \vec{H} = \vec{J}$$

$$3. \nabla \cdot \vec{D} = 0$$

$$4. \nabla \cdot \vec{B} = 0$$

• Integral form:-

$$1. \oint \vec{E} \cdot d\vec{l} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

$$2. \oint \vec{H} \cdot d\vec{l} = I = \int_S \vec{J} \cdot d\vec{S}$$

$$3. \oint \vec{D} \cdot d\vec{S} = 0$$

$$4. \oint \vec{B} \cdot d\vec{S} = 0$$



d) Static field

• Point form:-

1. $\nabla \times \vec{E} = 0$... from Faraday's law

2. $\nabla \times \vec{H} = \vec{J}$... from ACL

3. $\nabla \cdot \vec{D} = \rho_v$... from Gauss law

4. $\nabla \cdot \vec{B} = 0$... from Gauss law

• Integral form:-

1. $\oint \vec{E} \cdot d\vec{l} = 0$... from Faraday's law

2. $\oint \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{S}$... from ACL

3. $\oint_S \vec{D} \cdot d\vec{S} = \int_V \rho_v dv$... from Gauss law

4. $\oint_S \vec{B} \cdot d\vec{S} = 0$... from Gauss law

• $\vec{H} = H_m e^{j(\omega t + \beta z)} \vec{a}_x$ in free space, find $\vec{E}$ using Maxwell's eqⁿ.

• Given:- $\vec{H} = H_m e^{j(\omega t + \beta z)} \vec{a}_x$, $\vec{E} = ?$

We know that,

$$\nabla \times \vec{H} = \frac{\partial}{\partial t} \vec{D}$$

$$D = \epsilon E$$

$$\therefore \nabla \times \vec{H} = \frac{\partial}{\partial t} \epsilon \vec{E} \Rightarrow \nabla \times \vec{H} = \epsilon \frac{\partial}{\partial t} \vec{E} \quad \text{--- (1)}$$



$$\nabla \times \vec{H} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_m e^{j(\omega t + \beta z)} & 0 & 0 \end{vmatrix}$$

$$\nabla \times \vec{H} = -\vec{a}_y [-H_m e^{j(\omega t + \beta z)} \cdot j\beta]$$

$$\therefore \nabla \times \vec{H} = H_m e^{j(\omega t + \beta z)} \cdot j\beta \vec{a}_y \quad \text{--- (2)}$$

put (2) in (1)

$$\therefore H_m e^{j(\omega t + \beta z)} \cdot j\beta \vec{a}_y = \epsilon \frac{\partial}{\partial t} \vec{E}$$

integrate b.s. w.r.t t ,

$$\int H_m e^{j(\omega t + \beta z)} \cdot j\beta \vec{a}_y dt = \int \epsilon \frac{\partial}{\partial t} \vec{E} dt$$

$$\therefore \frac{j\beta H_m e^{j(\omega t + \beta z)}}{j\omega} \vec{a}_y = \epsilon \vec{E}$$

$$\therefore \frac{\beta}{\omega} H_m e^{j(\omega t + \beta z)} \vec{a}_y = \epsilon \vec{E}$$

$$\therefore \boxed{\vec{E} = \frac{\beta}{\epsilon \omega} H_m e^{j(\omega t + \beta z)} \vec{a}_y} \quad \text{V/m}$$

Timely
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$$\vec{E} \cdot \vec{G} = \vec{H} \times \vec{\nabla} \epsilon$$

$$\vec{G} = \vec{\nabla} \times \vec{H}$$



Assignment No: 3

A unit vector directed from region 1 to region 2 at planar boundary betⁿ two perfect dielectrics is given as $\bar{a}_{n12} = -\frac{2}{7}\bar{a}_x + \frac{3}{7}\bar{a}_y + \frac{6}{7}\bar{a}_z$. Assume $\epsilon_{r1} = 3$, $\epsilon_{r2} = 2$ & $\bar{E}_1 = 100\bar{a}_x + 80\bar{a}_y + 60\bar{a}_z$ V/m. Find $\bar{E}_2$.

Given: -

$$\bar{E}_1 = 100\bar{a}_x + 80\bar{a}_y + 60\bar{a}_z \text{ V/m } \epsilon_{r2} = 2$$

$$\bar{a}_{n12} = -\frac{2}{7}\bar{a}_x + \frac{3}{7}\bar{a}_y + \frac{6}{7}\bar{a}_z$$

$$\epsilon_{r1} = 3, \epsilon_{r2} = 2.$$

$$\epsilon_{r1} = 3$$

We know that,

$$\begin{aligned} \bar{E}_{in} &= (\bar{E}_1 \cdot \bar{a}_{n12}) \cdot \bar{a}_{n12} \\ &= \left[(100\bar{a}_x + 80\bar{a}_y + 60\bar{a}_z) \cdot \left(-\frac{2}{7}\bar{a}_x + \frac{3}{7}\bar{a}_y + \frac{6}{7}\bar{a}_z \right) \right] \cdot \left(-\frac{2}{7}\bar{a}_x + \frac{3}{7}\bar{a}_y + \frac{6}{7}\bar{a}_z \right) \end{aligned}$$

$$= \left[-\frac{200}{7} + \frac{240}{7} + \frac{360}{7} \right] \cdot \left(-\frac{2}{7}\bar{a}_x + \frac{3}{7}\bar{a}_y + \frac{6}{7}\bar{a}_z \right)$$

$$\bar{E}_{in} = 57.142 \left(-\frac{2}{7}\bar{a}_x + \frac{3}{7}\bar{a}_y + \frac{6}{7}\bar{a}_z \right)$$

$$\bar{E}_{in} = -16.32\bar{a}_x + 24.48\bar{a}_y + 48.97\bar{a}_z \text{ V/m}$$

We know that,

$$\bar{E}_{it} = \bar{E}_1 - \bar{E}_{in}$$

$$= (100\bar{a}_x + 80\bar{a}_y + 60\bar{a}_z) - (-16.32\bar{a}_x + 24.48\bar{a}_y + 48.97\bar{a}_z)$$

$$\bar{E}_{it} = 116.32\bar{a}_x + 55.52\bar{a}_y + 11.03\bar{a}_z \text{ V/m}$$

We know that,

$$\bar{E}_{it} = \bar{E}_{2t} \therefore \bar{E}_{2t} = 116.32\bar{a}_x + 55.52\bar{a}_y + 11.03\bar{a}_z \text{ V/m}$$



$$\frac{E_{1n}}{E_{2n}} = \frac{\epsilon_2}{\epsilon_1}$$

$$\therefore \bar{E}_{2n} = \frac{\epsilon_1}{\epsilon_2} \bar{E}_{1n}$$

$$= \frac{3}{2} (-16.32 \bar{a}_x + 24.48 \bar{a}_y + 48.97 \bar{a}_z)$$

$$\therefore \bar{E}_{2n} = -24.48 \bar{a}_x + 36.72 \bar{a}_y + 73.455 \bar{a}_z$$

$$\bar{E}_2 = \bar{E}_{2n} + \bar{E}_{2t}$$

$$= (-24.48 \bar{a}_x + 36.72 \bar{a}_y + 73.455 \bar{a}_z) + (116.32 \bar{a}_x + 55.52 \bar{a}_y + 11.03 \bar{a}_z)$$

$$\therefore \bar{E}_2 = 91.84 \bar{a}_x + 92.24 \bar{a}_y + 84.485 \bar{a}_z \text{ V}$$

2. Region 1 & 2 interface each other $z=0$ p. Region 1 has $\mu_{r1}=1.5$, region 2 has $\mu_{r2}=1$. The flux density, $\bar{B}_1 = 1.2 \bar{a}_x + 0.8 \bar{a}_y + 0.4 \bar{a}_z$ T is incident at boundary from region 1. Calculate $\bar{B}_2$, angles made by fields with the normal & interface by applying boundary condⁿ. Verify the angles computed with the help of Snell's law.

→ Given: $\bar{B}_1 = 1.2 \bar{a}_x + 0.8 \bar{a}_y + 0.4 \bar{a}_z$ T

$\mu_{r1}=1.5$, $\mu_{r2}=1$.

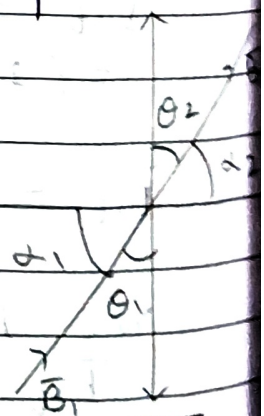
We know that,

$$\bar{B}_1 = \bar{B}_{n1} + \bar{B}_{t1}$$

$$\bar{B}_{n1} = 0.4 \bar{a}_z \text{ T}$$

$$\bar{B}_{t1} = 1.2 \bar{a}_x + 0.8 \bar{a}_y \text{ T, } \mu_{r1}=1.5$$

$$B_{n1} = B_{n2} = 0.4 \bar{a}_z$$





$$\frac{B_{t1}}{B_{t2}} = \frac{\mu_{r1}}{\mu_{r2}}$$

$$\bar{B}_{t2} = \frac{\mu_{r2}}{\mu_{r1}} \bar{B}_{t1}$$

$$= \frac{1}{1.5} [1.2\bar{a}_x + 0.8\bar{a}_y]$$

$$\bar{B}_{t2} = 0.8\bar{a}_x + 0.533\bar{a}_y \text{ T}$$

$$\therefore \bar{B}_2 = \bar{B}_{n2} + \bar{B}_{t2}$$

$$\bar{B}_2 = 0.8\bar{a}_x + 0.533\bar{a}_y + 0.4\bar{a}_z \text{ T}$$

We know that,

$$|\bar{B}_{n1}| = |\bar{B}_1| \cos \theta_1$$

$$|\bar{B}_{n1}| = 0.4, \quad |\bar{B}_1| = 1.496$$

$$\therefore B_{n1} = B_1 \cos \theta_1$$

$$0.4 = 1.496 \cos \theta_1$$

$$\cos \theta_1 = \frac{0.4}{1.496}$$

$$\theta_1 = \cos^{-1} \left(\frac{0.4}{1.496} \right)$$

$$\therefore \theta_1 = 74.49^\circ$$

$$\alpha_1 = 90 - \theta_1$$

$$= 90 - 74.49$$

$$\boxed{\alpha_1 = 15.51^\circ}$$

By,

$$B_{n2} = B_2 \cos \theta_2$$

$$|\bar{B}_{n2}| = 0.4, \quad |\bar{B}_2| = 1.041$$

$$\therefore B_{n2} = B_2 \cos \theta_2$$

$$0.4 = 1.041 \cos \theta_2$$



$$\therefore \cos \theta_2 = \frac{0.4}{1.041}$$

$$\therefore \theta_2 = \cos^{-1} \left(\frac{0.4}{1.041} \right)$$

$$\boxed{\therefore \theta_2 = 67.4^\circ}$$

$$\begin{aligned} \alpha_2 &= 90 - \theta_2 \\ &= 90 - 67.4 \end{aligned}$$

$$\boxed{\alpha_2 = 22.6^\circ}$$

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\tan (74.49)}{\tan (67.4)}$$

$$\frac{\tan \theta_1}{\tan \theta_2} = 1.5 \quad \text{--- (1)}$$

$$\frac{u_{r1}}{u_{r2}} = \frac{1.5}{1} = 1.5 \quad \text{--- (2)}$$

Compare (1) & (2)

$$\boxed{\therefore \frac{\tan \theta_1}{\tan \theta_2} = \frac{u_{r1}}{u_{r2}}}$$

... Snell's Law
verified.

Timely Submission	Understanding	Performance Presentation	Total
05	03	02	10/10